

Threefold Consensus-Driven Anchor Alignment for Efficient Multi-View Clustering

Chuan Tang , Miaomiao Li , Jun Wang , Renxiang Guan , Siwei Wang , *Member, IEEE*,
Chang Tang , *Senior Member, IEEE*, En Zhu , and Xinwang Liu , *Senior Member, IEEE*

Abstract—Anchor-based clustering methods have emerged as an effective paradigm for improving clustering efficiency, particularly in the context of large-scale data analysis. By introducing a compact set of representative samples, these methods significantly reduce computational complexity while preserving essential structural information within the original data. However, in multi-view scenarios, heterogeneous data distributions often lead to anchor misalignment, where anchors from different views fail to establish accurate correspondence. This Anchor Unalignment Problem (AUP) causes an incorrect graph fusion and disrupts the global structural consistency across views, ultimately degrading clustering performance. To this end, we propose a novel framework, named Threefold Consensus-Driven Anchor Alignment for Efficient Multi-View Clustering (TCAA), which jointly enforces representation consistency, spectral consistency, and discrete clustering consistency across views under an adaptive weighting scheme. Specifically, permutation matrices are employed to align anchor graphs across views under the guidance of representation similarity, while spectral structural signatures capture the global roles of anchors to ensure robust alignment. Furthermore, discrete clustering consistency directly couples the consensus anchor graph with cluster assignments, eliminating the information loss of two-stage methods. Extensive experiments on real-world datasets demonstrate that TCAA outperforms state-of-the-art multi-view clustering methods.

Index Terms—Multi-view clustering, large-scale clustering, anchor alignment, anchor graph learning.

I. INTRODUCTION

IN RECENT years, the rapid growth of multi-source and heterogeneous data has driven substantial progress in multi-view clustering (MVC), an unsupervised technique designed to group data by integrating complementary information from multiple perspectives or modalities. Unlike traditional single-view clustering methods, multi-view clustering methods simultaneously explore the consensus and diversity among different views to uncover the intrinsic structure of data more comprehensively [1], [2], [3], [4], [5]. By leveraging information from multiple perspectives, MVC has demonstrated superior clustering performance across various real-world scenarios, including image processing [6], [7], bioinformatics [8], natural language processing [9], social network analysis [10] and knowledge reasoning [11], [12]. However, despite its success in various applications, extending MVC to large-scale datasets remains a major challenge. The main difficulty lies in the substantial computational burden associated with scaling multi-view clustering methods to large-scale datasets. Operations such as constructing similarity graphs in graph-based clustering [13], [14], computing spectral embeddings in spectral clustering [15], [16], and optimizing self-representation matrices in subspace clustering [17], [18] are inherently computationally intensive. The computational cost of these approaches typically grows quadratically $\mathcal{O}(n^2)$ or even cubically $\mathcal{O}(n^3)$ with respect to the number of samples n , which makes conventional MVC methods impractical for large-scale scenarios.

To address the challenges of applying MVC methods to large-scale datasets, a variety of anchor-based approaches have been proposed [19], [20], [21], [22]. Specifically, instead of constructing the conventional complete $n \times n$ similarity graph, an $n \times m$ ($m \ll n$) anchor graph is constructed to capture the pairwise relationships between the entire set of samples and the selected representative anchors. Owing to the significantly smaller number of anchors, this surrogate representation markedly reduces both computational and memory demands. Therefore, anchor-based methods provide an effective and scalable paradigm for extending graph-based clustering techniques to large and complex multi-view datasets.

Although anchor-based learning strategies have achieved promising performance in multi-view clustering scenarios, they inevitably suffer from the anchor-unaligned problem (AUP), as illustrated in Fig. 1. Each view independently selects anchors and builds an anchor graph linking samples to their nearest anchors,

Received 22 October 2025; revised 30 March 2026; accepted 27 June 2026. Date of publication 6 July 2026; date of current version 8 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62476281, Grant 62325604, Grant 62276271, Grant 62406329, and Grant 62476280 and in part by the National Natural Science Foundation of China Joint Fund under Grant U24A20323. Recommended for acceptance by P. Wang. (Corresponding authors: Miaomiao Li; En Zhu.)

Chuan Tang, Jun Wang, Renxiang Guan, En Zhu, and Xinwang Liu are with the College of Computer Science and Technology, National University of Defense Technology, Changsha 410073, China (e-mail: tangchuan25@nudt.edu.cn; wang_jun@nudt.edu.cn; renxiangguan@nudt.edu.cn; enzhu@nudt.edu.cn; xinwangliu@nudt.edu.cn).

Miaomiao Li is with the College of Electronic Information and Electrical Engineering, Changsha University, Changsha 410022, China (e-mail: miaomiao.linudt@gmail.com).

Siwei Wang is with the Intelligent Game and Decision Lab, Academy of Military Sciences, 100091, China (e-mail: wangsiwei13@nudt.edu.cn).

Chang Tang is with the School of Software Engineering, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: tangchang@hust.edu.cn).

The code is publicly available at <https://github.com/tangchuan2000/TCAA>. Digital Object Identifier 10.1109/TKDE.2026.3710572

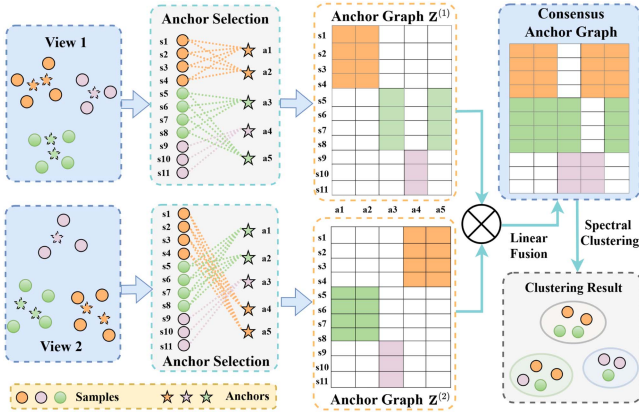


Fig. 1. Illustration of Anchor-Unaligned Problem. Direct fusion of anchor graphs through linear combination yields a consensus anchor graph with misaligned structures, where information from different clusters becomes intermixed. Performing spectral clustering on such an inconsistent graph results in degraded performance.

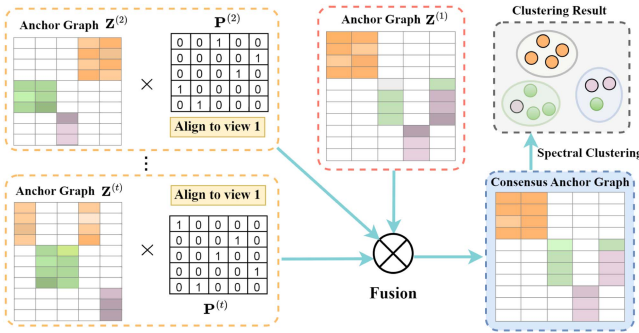


Fig. 2. The existing anchor alignment mechanism. These methods typically select a specific view as the reference and utilize permutation matrices to align the anchor graphs of the remaining views to it.

resulting in anchor graphs $Z^{(1)}$ and $Z^{(2)}$. Because the anchors are chosen separately in each view, their semantic meaning and distribution are not consistent, which leads to discrepancies in the anchor graphs. When these graphs are directly fused through linear combination to form a consensus anchor graph, the misalignment introduces mismatched structures that mix information from different clusters. Spectral clustering performed on this inconsistent consensus graph produces degraded clustering results, where samples from distinct clusters are incorrectly grouped.

To address the above challenge, conventional approaches typically select the anchor graph from one specific view as a reference and employ permutation matrices to align the anchor graphs of the remaining views to it, as illustrated in Fig. 2. In recent years, several pioneering studies have been proposed following this paradigm. Wang et al. [21] propose a FMVACC module that mitigates anchor inconsistency by introducing permutation matrices to align anchors from all views to a fixed reference view, enforcing both feature and structural correspondences before graph fusion. However, its reliance on the first view as the baseline may bias the results if that view is of low quality.

To overcome this drawback, Ma et al. [23] adopts an adaptive baseline selection strategy, where anchors are first learned in each view, and the view with the highest anchor quality, measured by a compactness criterion, is automatically chosen as the reference for alignment. More recently, Liu et al. [24] introduced an AMCA2 model that extends the scope of anchor alignment by integrating both anchor-wise and class-wise correspondences into a unified optimization framework, which simultaneously learns permutation matrices for one-to-one anchor matching and emphasizes discriminative anchors through a Hadamard product. Complementary to these efforts, the MV-CAGAF framework [1] addresses the Anchor-Unaligned Problem from the perspective of structural representation matching. Instead of relying on raw feature representations, it treats each anchor as an n -dimensional structural vector derived from the anchor graph, and formulates cross-view alignment as an optimal transport problem that minimizes the Earth Mover's Distance between anchor distributions.

While the above-mentioned methods have shown remarkable effectiveness in addressing the AUP problem, we observe that they still suffer from several inherent limitations. First, many approaches either place excessive emphasis on a single view [21], [23] or completely neglect inter-view differences during the alignment process [1], [24], thereby failing to fully exploit the complementary characteristics of multi-view information. Second, these methods often do not adequately utilize the structural properties of anchors: some disregard them entirely [23], [24], while others overemphasize structural comparisons between all features of anchors, which increases model complexity and reduces robustness [21]. Third, these methods require performing spectral clustering on the fused graph after the optimization stage. This two-step procedure may lead to information loss between the optimization and clustering phases, ultimately degrading overall clustering performance.

To this end, we propose a novel framework, termed Threefold Consensus-Driven Anchor Alignment (TCAA) for Efficient Multi-View Clustering, as illustrated in Fig. 3. Unlike approaches that align all views to a fixed reference view, TCAA does not rely on any single view as the benchmark. Instead, it employs adaptive view weights to balance contributions across views and jointly learns an aligned consensus anchor graph that integrates both representation and spectral consistency of anchors. Through iterative updates of alignment matrices and view weights, the model suppresses the influence of noisy or unreliable views while amplifying the contribution of informative ones, thereby eliminating reference-view bias and achieving more robust alignment. Moreover, clustering consistency is incorporated by directly aligning the consensus anchor graph with discrete cluster assignments, which unifies graph learning and clustering within a single framework, reduces information loss compared with two-stage methods, and enhances the discriminative capability of the learned consensus anchor graph.

The main contributions of our work are summarized as follows:

- We use adaptive view weights to balance contributions and jointly learn an aligned consensus anchor graph via the Hungarian algorithm, integrating both representation

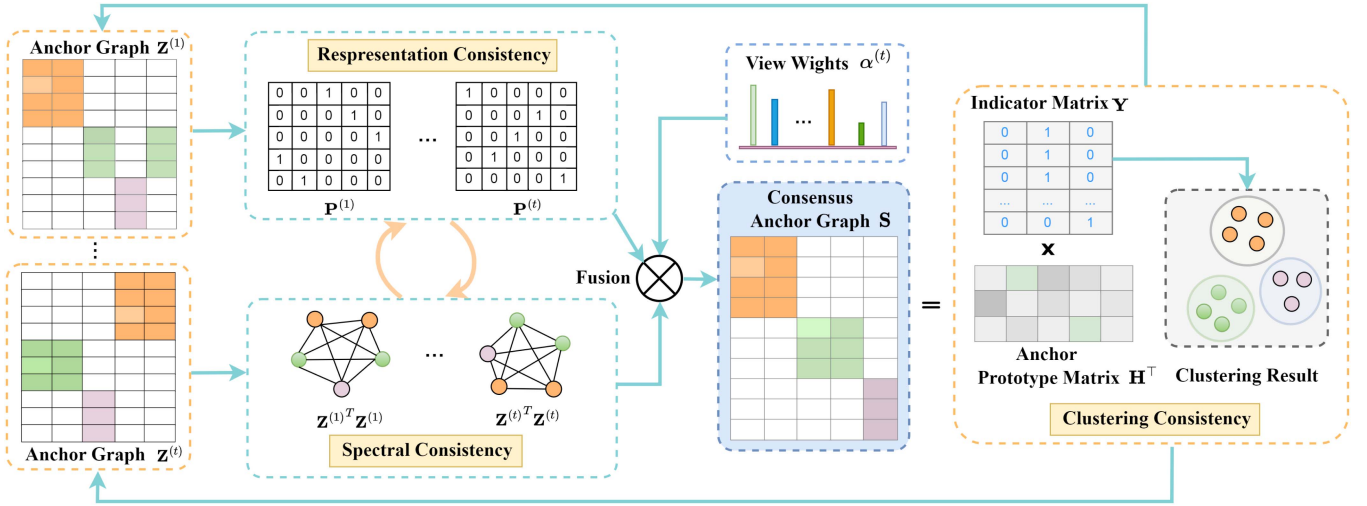


Fig. 3. Flowchart of our TCAA model. It adaptively balances the contributions of multiple views and jointly learns an aligned consensus anchor graph that integrates both representation and spectral consistencies. Moreover, clustering consistency is enforced by directly aligning the consensus anchor graph with discrete cluster assignments, thereby unifying graph learning and clustering within a single framework.

and spectral consistency. Unlike existing methods, our approach avoids reliance on any single view, ensuring robust cross-view alignment.

- We incorporate a discrete clustering consistency term to directly align the consensus anchor graph with cluster assignments, unifying graph learning and clustering. This approach mitigates the information loss of two-stage methods and strengthens the discriminative ability of the consensus representation.
- We develop an alternating optimization algorithm to effectively solve the resulting optimization problem. Extensive experimental evaluations demonstrate the superiority of our method in terms of both clustering performance and computational efficiency.

II. RELATED WORK

In this section, we first present the mathematical symbols in Table I. Then, we introduce several key concepts related to our proposed method, including the anchor selection strategy, anchor graph learning, and fusion, respectively.

A. Anchor Selection

A crucial issue in the anchor-based clustering method is how to effectively select the anchors from the whole dataset. Generally speaking, existing anchor selection strategies can be divided into two categories. The first is predefined anchor selection, in which anchors are determined before the clustering process. Typical techniques in this category include random sampling [25], [26], [27], directly alternate sampling (DAS) [28], [29], k -means [30], [31], [32] and BKHK [33], [34], all of which aim to capture representative samples from the original data distribution in a relatively simple and efficient manner. The second category is adaptive anchor learning, where anchor points are iteratively updated and refined during the optimization process [1], [35], [36], [37]. This dynamic strategy allows the

TABLE I
THE DEFINITION OF SYMBOLS

Symbols	Definition
v	number of views
m	number of anchors
n	number of samples
c	number of clusters
d_t	dimension of t -th view
l	dimension of anchor matrix
$\mathbf{I}$	identity matrix
$\mathbf{Y}^\top$	transpose of matrix $\mathbf{Y}$
$\text{Tr}(\mathbf{Y})$	trace of $\mathbf{Y}$
$\mathbf{Y}_{(i,j)}$	(i,j) -th element of $\mathbf{Y}$
$\mathbf{Y}_{i,:}$	i -th row of $\mathbf{Y}$
$\mathbf{Y}_{:,j}$	j -th column of $\mathbf{Y}$
$\ \cdot\ _F$	Frobenius norm

anchor set to progressively adapt to the intrinsic structure of the data, thereby improving representation quality and clustering performance. However, such adaptivity often comes at the expense of increased model complexity and higher computational overhead.

B. Anchor Graph Learning

After selecting anchors, the next step is to construct anchor graphs that represent the relationships between samples and anchors. For a multi-view dataset $\{\mathbf{X}^{(t)} \in \mathbb{R}^{n \times d_t}\}_{t=1}^v$, one approach to anchor graph learning can be expressed as follows:

$$\begin{aligned} \min_{\mathbf{Z}^{(t)}} \sum_{t=1}^v \left\| \mathbf{X}^{(t)} - \mathbf{Z}^{(t)} \mathbf{A}^{(t)} \right\|_F^2 + \alpha \mathcal{F}(\mathbf{Z}^{(t)}), \\ \text{s.t. } \mathbf{Z}^{(t)} \mathbf{1} = \mathbf{1}, \mathbf{Z}^{(t)} \geq 0, \end{aligned} \quad (1)$$

where $\mathbf{A}^{(t)} \in \mathbb{R}^{m \times d_t}$ represents the anchor matrix, which is learned through a certain anchor selection strategy. $\mathbf{Z}^{(t)} \in \mathbb{R}^{n \times m}$ represents the specific anchor graph that denotes the similarity relationship between the original data and the selected anchors in the t -th view. Specifically, each element $\mathbf{Z}_{(i,j)}^{(t)}$ represents the similarity between the i -th sample and the j -th anchor in the t -th view. The larger $\mathbf{Z}_{(i,j)}^{(t)}$ is, the higher the similarity between the i -th sample and the j -th anchor. By reducing the size of the graph matrix from $n \times n$ to $n \times m$, the computation complexity is reduced linearly to n . $\mathcal{F}(\cdot)$ represents a regularization function. Common ways include the Frobenius norm [38], [39], [40], $l_{1,2}$ norm [41] and tensor low-rank constraint [42]. Although this adaptive learning method is capable of producing sparse anchor graphs that better capture local structural relationships, its adaptive nature inevitably increases computational overhead.

Another approach to constructing the anchor graph is based on the similarities between the samples and the selected anchors [43], [44], and it can be formulated as follows:

$$\text{sim}(x_i^{(t)}, a_j^{(t)}) = \exp\left(-\frac{\|x_i^{(t)} - a_j^{(t)}\|^2}{2\sigma^{(t)^2}}\right), \quad (2)$$

where $x_i^{(t)}$ and $a_j^{(t)}$ denote the i -th sample and j -th anchor in the t -th view, respectively. $\sigma^{(t)}$ is the bandwidth parameter, typically set to the mean of all sample-anchor distances, i.e., $\sigma^{(t)} = \text{mean}_{i,j} \|x_i^{(t)} - a_j^{(t)}\|_2$. To achieve a sparse and locally consistent representation, each sample is connected only to its k nearest anchors, with the remaining connections set to zero. The similarities between the sample and its associated anchors are subsequently normalized as:

$$\mathbf{Z}_{ij}^{(t)} = \begin{cases} \frac{\text{sim}(x_i^{(t)}, a_j^{(t)})}{\sum_{j' \in \mathcal{N}_k(i)} \text{sim}(x_i^{(t)}, a_{j'}^{(t)})}, & j \in \mathcal{N}_k(i), \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

where $\mathcal{N}_k(i)$ denotes the set of the k nearest anchors of sample i . In practice, k is often chosen as 3, 5, or 15. Following [45], we set $k = 3$ in our proposed method.

The use of anchors reduces the complexity of graph construction from $\mathcal{O}(n^2)$ (full sample-sample graphs) to $\mathcal{O}(nm)$ with $m \ll n$, making the method scalable to large datasets.

C. Anchor Graph Fusion

After obtaining the anchor graphs for all views, the next step is to adopt appropriate methods to fuse them. One common approach is direct concatenation [38], where the anchor graphs from all views are simply concatenated into a single large graph, as shown by $\mathbf{Z} = [\mathbf{Z}^{(1)}, \mathbf{Z}^{(2)}, \dots, \mathbf{Z}^{(t)}]$. Another method is linear fusion [46], which sums the anchor graphs from different views to obtain a consensus graph, represented as $\mathbf{Z} = \sum_{t=1}^v \alpha^{(t)} \mathbf{Z}^{(t)}$, where $\alpha^{(t)}$ is the weight of the t -th view. A more refined approach involves minimizing the Frobenius norm, where the objective is to minimize the difference between the consensus graph $\mathbf{Z}$ and the anchor graphs $\mathbf{Z}^{(t)}$ of each view, as shown by the formula $\sum_{t=1}^v \alpha^{(t)} \|\mathbf{Z} - \mathbf{Z}^{(t)}\|_F^2$ [40]

or $\sum_{t=1}^v \|\mathbf{Z} - \alpha^{(t)} \mathbf{Z}^{(t)}\|_F^2$ [28], where $\alpha^{(t)}$ is the weight of the t -th view. These methods aim to integrate complementary information from multiple views to generate a more accurate and robust anchor graph, thereby enhancing overall clustering performance. However, a critical issue is that they do not account for the Anchor-Unaligned Problem, leading to a suboptimal consensus anchor graph and adversely affecting the clustering results.

III. METHOD

In this section, we begin by introducing each part of the proposed model, followed by a detailed explanation of the optimization procedure and complexity analysis. Finally, we delve into the complexity analysis of TCAA.

A. Representation Consistency

As previously mentioned, $\mathbf{Z}^{(t)} \in \mathbb{R}^{n \times m}$ represents the specific anchor graph that denotes the similarity relationship between the original data and the selected anchors in the t -th view. Specifically, the values in each column of $\mathbf{Z}^{(t)}$ can be interpreted as the probability of each sample being connected to the corresponding anchor. For instance, in $\mathbf{Z}^{(1)}$ shown in Fig. 1, the probabilities of anchors $a1$ and $a2$ being connected to samples $s1$, $s2$, $s3$, and $s4$ are very high. Therefore, based on the principle of data consistency across different views in multi-view data, in $\mathbf{Z}^{(2)}$, there should also be anchors connected to samples $s1$, $s2$, $s3$, and $s4$, with similarly high probability values. Under this assumption, we can permute the columns of $\mathbf{Z}^{(2)}$ to obtain an aligned anchor graph that is consistent with $\mathbf{Z}^{(1)}$ [21], which can be formulated as:

$$\begin{aligned} \min_{\mathbf{P}} \quad & \|\mathbf{Z}^{(2)} \mathbf{P} - \mathbf{Z}^{(1)}\|_F^2, \\ \text{s.t.} \quad & \mathbf{P} \mathbf{1} = \mathbf{1}, \mathbf{P}^\top \mathbf{1} = \mathbf{1}, \mathbf{P} \in \{0, 1\}^{m \times m}, \end{aligned} \quad (4)$$

where $\mathbf{P}$ is a permutation matrix. This method suffers from a major limitation: when more than two views are involved, the permutation process is performed solely with respect to the first view. Consequently, if the first view is of low quality or contains noise, the resulting aligned consensus graph may become suboptimal. To overcome this drawback, we introduce a dynamic fusion strategy that adaptively learns a consensus anchor graph, formulated as follows:

$$\begin{aligned} \min_{\alpha^{(t)} \mathbf{P}^{(t)}, \mathbf{S}} \quad & \sum_{t=1}^v \alpha^{(t)^2} \|\mathbf{Z}^{(t)} \mathbf{P}^{(t)} - \mathbf{S}\|_F^2, \\ \text{s.t.} \quad & \mathbf{P}^{(t)} \mathbf{1} = \mathbf{1}, \mathbf{P}^{(t)\top} \mathbf{1} = \mathbf{1}, \mathbf{P}^{(t)} \in \{0, 1\}^{m \times m}, \\ & \mathbf{S} \mathbf{1} = \mathbf{1}, \mathbf{S} \geq 0, \sum_{t=1}^v \alpha^{(t)} = 1, \alpha^{(t)} \geq 0, \end{aligned} \quad (5)$$

where $\mathbf{P}^{(t)}$ is a permutation matrix for the t -th view. $\mathbf{S}$ represents the consensus anchor graph, which is the final fused anchor graph obtained after aligning anchors from all views. $\alpha^{(t)}$ is the weight coefficient that balances the influence of each view. Eq (5)

ensures that the anchor representation guides the alignment in each view.

B. Spectral Consistency

The representation consistency ensures that the anchor graph $\mathbf{Z}^{(t)}$ from each view is aligned with the consensus graph $\mathbf{S}$ based on the sample-to-anchor similarities. However, relying solely on representation consistency for aligning anchors might pair two representation-similar anchors together, even if they serve different roles in the graph structure in other views. For example, two anchors might have similar sample-to-anchor similarities in a particular view but represent different cluster roles, such as a cluster center and a cluster boundary. Spectral consistency addresses this issue by ensuring that anchors with similar global roles in the graph are aligned across views, complementing representation consistency, which might fail in cases where local similarity does not match the global structure. The spectral consistency can be formulated as:

$$\begin{aligned} \min_{\alpha^{(t)}, \mathbf{P}^{(t)}, \mathbf{Q}} \sum_{t=1}^v \alpha^{(t)2} \left\| \mathbf{P}^{(t)\top} \mathbf{Q}^{(t)} - \mathbf{Q} \right\|_F^2, \\ \text{s.t. } \mathbf{P}^{(t)} \mathbf{1} = \mathbf{1}, \mathbf{P}^{(t)\top} \mathbf{1} = \mathbf{1}, \mathbf{P}^{(t)} \in \{0, 1\}^{m \times m}, \\ \mathbf{Q}^\top \mathbf{Q} = \mathbf{I}, \sum_{t=1}^v \alpha^{(t)} = 1, \alpha^{(t)} \geq 0, \end{aligned} \quad (6)$$

where $\mathbf{Q} \in \mathbb{R}^{m \times c}$ denotes the consensus spectral signature matrix, which serves as a shared structural representation of anchors across all views. $\mathbf{Q}^{(t)}$ represents the spectral signatures of anchors for the t -th view, where each row corresponds to an anchor and each column denotes one of the top- c eigenvector dimensions. By comparing $(\mathbf{P}^{(t)})^\top \mathbf{Q}^{(t)}$ with the consensus spectral signature matrix $\mathbf{Q}$, the model enforces anchors from different views to be aligned not only by local similarities but also by their structural roles. The specific steps for calculating $\mathbf{Q}^{(t)}$ is as follows: Firstly, we compute the anchor-anchor affinity matrix $\mathbf{G}^{(t)}$ as $\mathbf{G}^{(t)} = \text{norm}(\mathbf{Z}^{(t)\top} \mathbf{Z}^{(t)}) = \mathbf{D}^{(t)(-\frac{1}{2})} (\mathbf{Z}^{(t)\top} \mathbf{Z}^{(t)}) \mathbf{D}^{(t)(-\frac{1}{2})} \in \mathbb{R}^{m \times m}$, where $\mathbf{D}^{(t)}$ is the diagonal degree matrix with entries $\mathbf{D}_{(i,i)}^{(t)} = \sum_j (\mathbf{Z}^{(t)\top} \mathbf{Z}^{(t)})_{i,j}$. Next, we perform eigen-decomposition on $\mathbf{G}^{(t)}$, i.e. $\mathbf{G}^{(t)} = \mathbf{U}^{(t)} \Sigma^{(t)} (\mathbf{U}^{(t)})^\top$, where $\mathbf{U} \in \mathbb{R}^{m \times m}$ is the orthogonal eigenvector matrix, and $\Sigma \in \mathbb{R}^{m \times m}$ is the diagonal matrix of eigenvalues. The columns of $\mathbf{U}^{(t)}$ correspond to the spectral embeddings of anchors in the t -th view. In practice, only the top- c eigenvectors associated with the largest eigenvalues are retained, yielding the spectral signature matrix $\mathbf{Q}^{(t)} = \mathbf{U}_{(:,1:c)}^{(t)}$. The rationale for constructing $\mathbf{Q}^{(t)}$ from the leading c eigenvectors is that they capture the most informative structural directions corresponding to the c clusters, while discarding higher-order eigenvectors helps filter out noise and suppress the influence of outliers. As a result, we obtain a compact and robust spectral signature space that facilitates efficient and accurate anchor alignment across views.

C. Discrete Clustering Consistency

In many anchor-based clustering methods, once the consensus anchor graph is generated, it is common to perform an SVD followed by k -means clustering [47], [48], [49], [50]. These two-stage strategies often suffer from information loss by separating graph construction and clustering. To overcome this, we introduce a discrete clustering consistency term that directly couples the consensus anchor graph with cluster assignments. This enables end-to-end optimization and yields a more discriminative consensus representation, which can be formulated as:

$$\min_{\mathbf{Y}, \mathbf{H}} \left\| \mathbf{S} - \mathbf{Y} \mathbf{H}^\top \right\|_F^2, \text{ s.t. } \mathbf{Y} \mathbf{1} = \mathbf{1}, \mathbf{Y} \in \{0, 1\}^{n \times c}, \quad (7)$$

where $\mathbf{Y}$ denotes the cluster indicator matrix, with $\mathbf{Y}_{(i,j)} = 1$ if sample i belongs to cluster j and $\mathbf{Y}_{(i,j)} = 0$ otherwise. $\mathbf{H} \in \mathbb{R}^{m \times c}$ is the anchor prototype matrix, where each column $\mathbf{H}_{:,j}$ corresponds to the centroid of cluster j in the anchor space, serving as a representative embedding that captures the structural characteristics of the cluster.

D. Formulation of TCAA

By integrating the above representation, spectral, and discrete clustering consistency, our TCAA model can be formulated as:

$$\begin{aligned} \min_{\alpha^{(t)}, \mathbf{P}^{(t)}, \mathbf{S}, \mathbf{Q}, \mathbf{Y}, \mathbf{H}} \sum_{t=1}^v \alpha^{(t)2} \left(\left\| \mathbf{Z}^{(t)} \mathbf{P}^{(t)} - \mathbf{S} \right\|_F^2 \right. \\ \left. + \beta \left\| \mathbf{P}^{(t)\top} \mathbf{Q}^{(t)} - \mathbf{Q} \right\|_F^2 \right) + \lambda \left\| \mathbf{S} - \mathbf{Y} \mathbf{H}^\top \right\|_F^2, \\ \text{s.t. } \mathbf{P}^{(t)} \mathbf{1} = \mathbf{1}, \mathbf{P}^{(t)\top} \mathbf{1} = \mathbf{1}, \mathbf{P}^{(t)} \in \{0, 1\}^{m \times m}, \\ \mathbf{S} \mathbf{1} = \mathbf{1}, \mathbf{S} \geq 0, \mathbf{Q}^\top \mathbf{Q} = \mathbf{I}, \sum_{t=1}^v \alpha^{(t)} = 1, \alpha^{(t)} \geq 0, \\ \mathbf{Y} \mathbf{1} = \mathbf{1}, \mathbf{Y} \in \{0, 1\}^{n \times c}, \end{aligned} \quad (8)$$

where $\alpha^{(t)}$ denotes the weight coefficient that balances the contributions of different views, while β and λ are hyperparameters. Specifically, β regulates the trade-off between spectral consistency and representation consistency, whereas λ controls the strength of cluster consistency. The first term enforces representation consistency, aligning anchor graphs from different views with the consensus graph based on sample-anchor similarities. The second term introduces spectral consistency, which leverages spectral anchor signatures to capture the global structural roles of anchors, thus improving robustness against noise and mitigating the influence of ambiguous local similarities. The third term incorporates discrete clustering consistency, directly coupling the consensus anchor graph with cluster assignments to unify graph learning and clustering in an end-to-end manner. By jointly optimizing these three terms, the proposed model not only achieves precise anchor alignment across views but also learns discriminative consensus representations that enhance clustering performance.

E. Relationship Between TCAA and FMVACC

FMVACC [21] is a pioneering work that first investigates the AUP for anchor graph fusion tasks. It adopts a reference-view strategy, where all non-reference views are aligned to the first view through pairwise anchor matching, after which the fused anchor graph is constructed from the aligned anchors. In contrast, TCAA introduces a threefold consensus-driven alignment paradigm that eliminates reliance on any specific reference view. In particular, the spectral consistency component leverages spectral anchor signatures to capture the global structural roles of anchors, thereby enhancing robustness and mitigating noise. Moreover, TCAA incorporates adaptive view weighting, allowing informative views to contribute more while suppressing those that are noisy or less reliable. Finally, whereas FMVACC performs the “align-then-fuse” operation sequentially, TCAA accomplishes joint anchor alignment and fusion within a unified optimization framework, resulting in improved robustness and superior clustering performance.

IV. OPTIMIZATION

A. Optimization Algorithm

Since the optimization problem of (8) is not jointly convex for all variables $\mathbf{P}^{(t)}$, $\alpha^{(t)}$, $\mathbf{S}$, $\mathbf{Y}$, $\mathbf{Q}$ and $\mathbf{H}$, we decompose it into six subproblems for solution.

Sub-problem of $\mathbf{P}^{(t)}$: When fixing $\mathbf{S}$, $\alpha^{(t)}$, $\mathbf{Y}$, $\mathbf{Q}$ and $\mathbf{H}$, the optimization for $\mathbf{P}^{(t)}$ can be reformulated as:

$$\begin{aligned} \min_{\mathbf{P}^{(t)}} \sum_{t=1}^v \left\| \mathbf{Z}^{(t)} \mathbf{P}^{(t)} - \mathbf{S} \right\|_F^2 + \beta \left\| \mathbf{P}^{(t)\top} \mathbf{Q}^{(t)} - \mathbf{Q} \right\|_F^2, \\ \text{s.t. } \mathbf{P}^{(t)} \mathbf{1} = \mathbf{1}, \mathbf{P}^{(t)\top} \mathbf{1} = \mathbf{1}, \mathbf{P}^{(t)} \in \{0, 1\}^{m \times m}. \end{aligned} \quad (9)$$

This optimization problem can be formulated as a Linear Assignment Problem (LAP), where each anchor in view t must be assigned to one consensus anchor. The cost matrix for the LAP is constructed by combining representation-based and spectral-based dissimilarities. Specifically, the representation dissimilarity is defined as:

$$\mathbf{C}_{\text{representation}}^{(t)} = \left\| \mathbf{Z}_{:,i}^{(t)} - \mathbf{S}_{:,j} \right\|^2, \quad (10)$$

while the spectral dissimilarity is given by:

$$\mathbf{C}_{\text{spectral}}^{(t)} = \left\| \mathbf{Q}_{i,:}^{(t)} - \mathbf{Q}_{j,:} \right\|^2. \quad (11)$$

Thus, the overall cost matrix of the t -th view is expressed as:

$$\mathbf{C}^{(t)} = \mathbf{C}_{\text{representation}}^{(t)} + \beta \mathbf{C}_{\text{spectral}}^{(t)}. \quad (12)$$

Consequently, problem (9) is reduced to a standard LAP formulation:

$$\min_{\mathbf{P}^{(t)} \in \Pi_m} \langle \mathbf{C}^{(t)}, \mathbf{P}^{(t)} \rangle, \quad (13)$$

where $\Pi_m = \{\mathbf{P}^{(t)} \mathbf{1} = \mathbf{1}, \mathbf{P}^{(t)\top} \mathbf{1} = \mathbf{1}, \mathbf{P}^{(t)} \in \{0, 1\}^{m \times m}\}$. The standard LAP in (13) can be efficiently solved using the Hungarian algorithm [51], [52].

Sub-problem of $\mathbf{S}$: During the update of $\mathbf{S}$, (8) is equivalently converted into:

$$\begin{aligned} \min_{\mathbf{S}} \sum_{t=1}^v \alpha^{(t)^2} \left\| \mathbf{Z}^{(t)} \mathbf{P}^{(t)} - \mathbf{S} \right\|_F^2 + \lambda \left\| \mathbf{S} - \mathbf{Y} \mathbf{H}^\top \right\|_F^2, \\ \text{s.t. } \mathbf{S} \mathbf{1} = \mathbf{1}, \mathbf{S} \geq 0. \end{aligned} \quad (14)$$

Let

$$\Gamma = \sum_{t=1}^v \alpha^{(t)^2} + \lambda, \quad \hat{\mathbf{S}} = \frac{\sum_{t=1}^v \alpha^{(t)^2} \mathbf{Z}^{(t)} \mathbf{P}^{(t)} + \lambda \mathbf{Y} \mathbf{H}^\top}{\sum_{t=1}^v \alpha^{(t)^2} + \lambda},$$

then (14) can be rewritten as:

$$\min_{\mathbf{S}} \Gamma \left\| \mathbf{S} - \hat{\mathbf{S}} \right\|_F^2 + \mathbf{C}_{\text{cons}}, \quad (15)$$

where $\mathbf{C}_{\text{cons}} = \sum_{t=1}^v \alpha^{(t)^2} \left\| \mathbf{Z}^{(t)} \mathbf{P}^{(t)} \right\|_F^2 + \lambda \left\| \mathbf{Y} \mathbf{H}^\top \right\|_F^2 - \Gamma \left\| \hat{\mathbf{S}} \right\|_F^2$, which is a constant independent of $\mathbf{S}$. By ignoring this constant term, (15) reduces to:

$$\min_{\mathbf{S}} \Gamma \left\| \mathbf{S} - \hat{\mathbf{S}} \right\|_F^2. \quad (16)$$

The unconstrained optimal solution is given by $\mathbf{S}^* = \hat{\mathbf{S}}$. To satisfy the probability simplex constraints ($\mathbf{S} \mathbf{1} = \mathbf{1}, \mathbf{S} \geq 0$), each row of $\mathbf{S}^*$ is projected onto the simplex. This projection can be implemented by employing a dedicated simplex projection algorithm [53].

Sub-problem of $\mathbf{Q}$: When fixing α , $\mathbf{P}^{(t)}$, $\mathbf{S}$, $\mathbf{Y}$, and $\mathbf{H}$, (8) can be equivalently reformulated as:

$$\min_{\mathbf{Q}} \sum_{t=1}^v \alpha^{(t)^2} \left\| \mathbf{P}^{(t)\top} \mathbf{Q}^{(t)} - \mathbf{Q} \right\|_F^2 \text{ s.t. } \mathbf{Q}^\top \mathbf{Q} = \mathbf{I}. \quad (17)$$

First, the unconstrained optimal solution of $\mathbf{Q}$ is obtained as:

$$\mathbf{Q}^* = \frac{\sum_{t=1}^v \alpha^{(t)^2} \mathbf{P}^{(t)\top} \mathbf{Q}^{(t)}}{\sum_{t=1}^v \alpha^{(t)^2}}. \quad (18)$$

Subsequently, singular value decomposition (SVD) is performed on $\mathbf{Q}^*$, i.e., $\mathbf{Q}^* = \mathbf{U} \Sigma \mathbf{V}$. According to the orthogonality constraint in (17), the optimal solution is then given by $\mathbf{Q} = \mathbf{U} \mathbf{V}^\top$.

Sub-problem of $\mathbf{Y}$: Fixing other variables, the optimization problem (8) involving $\mathbf{Y}$ can be reformulated as:

$$\min_{\mathbf{Y}} \left\| \mathbf{S} - \mathbf{Y} \mathbf{H}^\top \right\|_F^2, \text{ s.t. } \mathbf{Y} \mathbf{1} = \mathbf{1}, \mathbf{Y} \in \{0, 1\}. \quad (19)$$

The problem reduces to assigning each sample to the cluster whose prototype provides the best reconstruction of its consensus anchor representation. More specifically, the optimization can be expressed row-wise as:

$$\begin{aligned} \mathbf{Y}_{i,:}^* &= \arg \min_{\mathbf{y}} \left\| \mathbf{S}_{i,:} - \mathbf{H} \mathbf{y} \right\|_2^2, \\ \text{s.t. } \mathbf{y} &\in \{0, 1\}^c, \mathbf{1}^\top \mathbf{y} = 1, \forall i \in \{1, \dots, n\}, \end{aligned} \quad (20)$$

where $\mathbf{y}$ represents the i -th row of $\mathbf{Y}$. Eq (20) is equivalent to a closest-prototype assignment:

$$\mathbf{Y}_{(i,k)}^* = \begin{cases} 1, & k = \arg \min_j \left\| \mathbf{S}_{i,:} - \mathbf{H}_{:,j} \right\|_2^2, \\ 0, & \text{otherwise.} \end{cases} \quad (21)$$

Algorithm 1: The Proposed TCAA.

Input: Data matrix $\mathbf{X}$, parameters β and λ , cluster number c .

- 1: Initialization: Set $\alpha^{(t)} = 1/v$. Initialize the indicator matrix $\mathbf{Y} \in \mathbb{R}^{n \times c}$ with one-hot rows, where the first c rows form an identity matrix and the remaining rows are assigned cyclically. Obtain the anchor matrices $\mathbf{A}^{(t)}$ using the k -means strategy. Construct the anchor graphs $\mathbf{Z}^{(t)}$ by (3). Compute $\mathbf{Q}^{(t)}$ as described in the previous subsection ‘Spectral Consistency’. Initialize the consensus graph $\mathbf{S}$ as the average of all $\mathbf{Z}^{(t)}$, $\epsilon = 1e - 4$.
- 2: **while** not converged **do**
- 3: Update $\mathbf{P}^{(t)}$ via (13).
- 4: Update $\mathbf{S}$ via (16).
- 5: Update $\mathbf{Q}$ via (18).
- 6: Update $\mathbf{Y}$ via (21).
- 7: Update $\mathbf{H}$ via (23).
- 8: Update $\alpha^{(t)}$ via (25).
- 9: Check convergence conditions:
 iteration times(r) ≥ 100 or
 $|\mathcal{L}^{(r+1)} - \mathcal{L}^{(r)}|/|\mathcal{L}^{(r)}| < \epsilon$.
- 10: **end while**

Output: Indicator matrix $\mathbf{Y}$.

Hence, each sample is assigned to the group whose anchor prototype $\mathbf{H}_{:,j}$ is closest to its consensus anchor representation.

Sub-problem of H: During the update of $\mathbf{H}$, (8) is equivalently converted into:

$$\min_{\mathbf{H}} \|\mathbf{S} - \mathbf{Y}\mathbf{H}^\top\|_F^2. \quad (22)$$

Expanding and taking the derivative with respect to $\mathbf{H}$ yields the closed-form solution:

$$\mathbf{H} = \mathbf{S}^\top \mathbf{Y} (\mathbf{Y}^\top \mathbf{Y} + \epsilon \mathbf{I})^{-1}, \quad (23)$$

where ϵ is a very small positive constant introduced for numerical stability and to avoid singularity of $\mathbf{Y}^\top \mathbf{Y}$.

Sub-problem of $\alpha^{(t)}$: When fixing $\mathbf{P}^{(t)}$, $\mathbf{S}$, $\mathbf{Y}$, and $\mathbf{H}$, (8) can be equivalently reformulated as:

$$\min_{\alpha^{(t)}} \sum_{t=1}^v \alpha^{(t)^2} \mathcal{E}^{(t)}, \text{ s.t. } \sum_{t=1}^v \alpha^{(t)} = 1, \alpha^{(t)} \geq 0, \quad (24)$$

where $\mathcal{E}^{(t)} = \|\mathbf{Z}^{(t)} \mathbf{P}^{(t)} - \mathbf{S}\|_F^2 + \beta \|(\mathbf{P}^{(t)})^\top \mathbf{Q}^{(t)} - \mathbf{Q}\|_F^2$. Applying the Karush-Kuhn-Tucker (KKT) conditions [54], the closed-form solution for each $\alpha^{(t)}$ is derived as:

$$\alpha^{(t)} = \frac{\frac{1}{\mathcal{E}^{(t)}}}{\sum_{j=1}^v \frac{1}{\mathcal{E}^{(j)}}}. \quad (25)$$

In summary, the strategy for solving problem (8) is presented in Algorithm 1.

TABLE II
DETAILS OF MULTI-VIEW DATASETS IN OUR EXPERIMENTS

Datasets	Abbr	#Views	#Samples	#Clusters
COIL20	COIL20	3	1440	20
Caltech101-20	Cal20	6	2386	20
YouTubeFace10	YTF10	4	38654	10
MNIST	MNIST	6	60000	10
Cifar10	Cifar10	4	60000	10
YouTubeFace20	YTF20	4	63896	20
YouTubeFace50	YTF50	4	126054	50

B. Complexity Analysis

The computational complexity of optimizing TCAA involves anchor selection, anchor graph construction, spectral signature computation, and updating six variables. Anchor selection using k -means requires $\mathcal{O}(nmd_t)$. Constructing the anchor graph $\mathbf{Z}^{(t)}$ entails computing similarities between n samples and m anchors, leading to a cost of $\mathcal{O}(nm)$. The computation of $\mathbf{Q}^{(t)}$ involves $\mathcal{O}(nm^2)$ for obtaining the anchor-anchor affinity matrix $\mathbf{G}^{(t)}$ and $\mathcal{O}(m^3)$ for eigen-decomposition. The update of permutation matrices $\mathbf{P}^{(t)}$ is formulated as a linear assignment problem, which can be efficiently solved by the Hungarian algorithm with complexity $\mathcal{O}(m^3)$. Updating $\mathbf{S}$ requires $\mathcal{O}(nm^2)$ to compute the unconstrained solution $\mathbf{S}^*$ and $\mathcal{O}(nm)$ for the simplex projection algorithm. The update of the cluster indicator matrix $\mathbf{Y}$ involves assigning each sample to its nearest prototype, incurring $\mathcal{O}(nmc)$. Solving $\mathbf{H}$ also costs $\mathcal{O}(nmc)$. Updating $\{\alpha^{(t)}\}$ is dominated by the computation of $\mathcal{E}^{(t)}$, with a cost of $\mathcal{O}(nm^2)$. Therefore, assuming that solving (8) requires r iterations, the total computational complexity is $\mathcal{O}(vnm d_t + vnm + vnm^2 + m^3 + r(vm^3 + nm^2 + nm + 2nmc + vnm^2))$. Given that $m \ll n$, $v \ll n$, and $c \ll n$, the overall complexity of TCAA reduces to $\mathcal{O}(rvnm^2)$, which is substantially more efficient than full graph-based multi-view clustering methods that typically incur $\mathcal{O}(rvn^3)$. This efficiency highlights the scalability of TCAA to large-scale multi-view datasets.

The memory consumption of TCAA is dominated by storing anchor graphs and consensus variables. For each view, the anchor graph $\mathbf{Z}^{(t)}$ requires $\mathcal{O}(nm)$ storage, and across all v views this totals $\mathcal{O}(vnm)$. The consensus anchor graph $\mathbf{S}$ also requires $\mathcal{O}(nm)$ memory. The permutation matrices $\mathbf{P}^{(t)} \in \{0, 1\}^{m \times m}$ consume $\mathcal{O}(vm^2)$, which is relatively small since $m \ll n$. The spectral signatures $\mathbf{Q}^{(t)} \in \mathbb{R}^{m \times c}$ and consensus signature matrix $\mathbf{Q} \in \mathbb{R}^{m \times c}$ each require $\mathcal{O}(mc)$. Finally, the cluster indicator matrix $\mathbf{Y} \in \{0, 1\}^{n \times c}$ and anchor prototypes $\mathbf{H} \in \mathbb{R}^{m \times c}$ require $\mathcal{O}(nc)$ and $\mathcal{O}(mc)$, respectively. Combining these terms, the overall space complexity can be summarized as: $\mathcal{O}(vnm + vm^2 + nm + nc + mc)$, where the dominant term is $\mathcal{O}(vnm)$. Compared with traditional multi-view clustering methods that need to construct and store full similarity graphs of size $\mathcal{O}(vn^2)$, the proposed approach dramatically reduces memory usage by leveraging anchor graphs, thereby enabling applications on large-scale datasets.

It is worth noting that the above analysis gives a conservative dense upper bound. In practice, each row of $\mathbf{Z}^{(t)}$ contains only k

nonzero entries because each sample is connected to its k nearest anchors only. Therefore, the storage cost of $\mathbf{Z}^{(t)}$ can be reduced from $\mathcal{O}(nm)$ to $\mathcal{O}(nk)$, and the computation of $(\mathbf{Z}^{(t)})^\top \mathbf{Z}^{(t)}$ can be implemented in $\mathcal{O}(nk^2)$ rather than the dense $\mathcal{O}(nm^2)$ form. Although some subsequent updates still involve dense matrices such as $\mathbf{S}$ and $\mathbf{YH}^\top$, this sparsity considerably reduces the practical runtime in anchor-graph-related computations.

V. EXPERIMENTS

A. Experimental Setup

Datasets: In our experiments, seven commonly used benchmark datasets, i.e., COIL20 [55], Caltech101-20 [48], MNIST [56], Cifar10 [57], YouTubeFace10,¹ YouTubeFace20¹, and YouTubeFace50¹ are used to evaluate the proposed method. Detailed information about them is presented in Table II.

Compared methods: We compared TCAA with the following multi-view methods: FMCNOF [46], EOMSC-CA [58], EMKMC [20], FMVACC [21], AEVC [59], 3AMVC [23], SMVAGC-SF [49], ALPC [60], OMVCDR [35].

Implementation details: To ensure a fair comparison, we fine-tuned the parameters of all baselines within the recommended ranges reported in their respective papers and executed each model five times. When a random seed was specified in the source code, the seed value was changed for each run to ensure variability. The clustering performance is reported as the mean and standard deviation across runs. For methods that require applying k -means on the consensus anchor graph (e.g., FMVACC, AEVC, 3AMVC, SMVAGC-SF, and ALPC), we performed k -means clustering 20 times to ensure result consistency. In our method, the number of anchors is selected from $\{1c, 2c, 4c, 5c, 10c\}$. Parameters β and λ are selected from $\{0.1, 1, 10, 100, 1000, 10000\}$, and $\{0.0001, 0.001, 0.01, 0.1, 1\}$, respectively. The clustering performance is evaluated using seven widely adopted metrics: Accuracy (ACC), Normalized Mutual Information (NMI), Adjusted Rand Index (ARI), F-score, Purity, Precision, and Recall. For all the above metrics, higher values indicate better clustering performance. All experiments are conducted on Windows 11 with an i9-14900 3.20 GHz CPU and 128 GB RAM.

B. Results Analysis

We evaluate the proposed method against nine representative algorithms on seven widely used benchmark datasets. The comparative results are reported in Table III, with the best performance highlighted in bold. Based on the results, several observations can be made.

- 1) Our model not only achieves competitive clustering results on small datasets (e.g., COIL20, Caltech101-20), but also demonstrates excellent performance on large-scale datasets (e.g., YouTubeFace10, MNIST, CIFAR10, YouTubeFace20, YouTubeFace50). In particular, with respect to ACC, our method consistently outperforms the second-best algorithm by 5.5%, 9.3%, 6.4%, 0.6%, 1.1%, 3.0%, and 3.2% on the corresponding datasets. In addition,

the proposed method also shows comparable or superior performance in other evaluation metrics. These promising results further support the effectiveness of the proposed method.

- 2) Compared with FMVACC and 3AMVC, which employ different anchor alignment strategies, our model achieves superior clustering performance. This advantage mainly stems from two key improvements. First, we adaptively learn an aligned and consistent anchor graph, instead of relying on the first view (i.e., FMVACC) or selecting a single optimal view as the benchmark (i.e., 3AMVC), since placing too much emphasis on one view may ignore the complementarity among multiple views. Second, by incorporating spectral consistency, we construct a compact and robust spectral signature space that facilitates efficient and accurate anchor alignment across views.
- 3) Compared with AEVC, SMVAGC-SF, and ALPC, which adopt a two-step procedure involving representation learning followed by k -means clustering, our model achieves superior results across all seven benchmark datasets. This demonstrates the effectiveness of the proposed one-step clustering strategy, which not only improves performance but also avoids the potential information loss inherent in multi-step clustering approaches.
- 4) In comparison to FMCNOF, EOMSC-CA, EMKMC, and OMVCDR, which are under a one-pass framework, TCAA also performs better. This advantage can be attributed to our threefold consensus-driven anchor alignment strategy, which learns an optimal consensus anchor graph to more accurately capture the interrelationships between anchors and samples, thereby uncovering the underlying relationships among samples. Moreover, TCAA consistently outperforms these one-step methods, demonstrating the effectiveness of the proposed one-step clustering strategy.

C. Running Time Comparison

To demonstrate the efficiency of the proposed algorithm, we record the execution time of TCAA and the compared methods on seven benchmark datasets in Fig. 4. Our model demonstrates higher efficiency compared to methods such as FMVACC, AEVC, 3AMVC, SMVAGC-SF, and ALPC, which require performing k -means many times on the learned consensus anchor graph. Another factor contributing to the efficiency of our model is the use of a Gaussian kernel when constructing the anchor graphs. This method produces only a small number of non-zero elements in each view's anchor graph, thereby reducing computational overhead. As a result, the optimization process of the model is also accelerated. While our method is not the fastest among all compared approaches, it still exhibits competitive efficiency, striking a good balance between performance and computational cost. This verifies its capability to effectively handle large-scale datasets.

D. Ablation Study

The strong performance of the proposed model can be largely attributed to the integration of three consistency strategies, each

¹<https://www.micc.unifi.it/resources/datasets/e-ytf/>

TABLE III
CLUSTERING PERFORMANCE ON SEVEN BENCHMARK DATASETS

Datasets	Methods	ACC	NMI	ARI	F-score	Purity	Precision	Recall
COIL20	FMCNOF	15.00 $\pm$ 0.46	32.20 $\pm$ 0.07	8.62 $\pm$ 0.78	16.65 $\pm$ 0.67	15.00 $\pm$ 1.90	9.35 $\pm$ 0.83	78.31 $\pm$ 0.45
	EOMSC-CA	92.57 $\pm$ 1.37	98.25 $\pm$ 1.46	93.57 $\pm$ 1.84	93.91 $\pm$ 1.74	95.01 $\pm$ 1.02	90.58 $\pm$ 0.54	97.47 $\pm$ 0.70
	FMVACC	90.38 $\pm$ 1.92	96.92 $\pm$ 0.00	90.16 $\pm$ 1.47	90.67 $\pm$ 0.66	92.35 $\pm$ 0.97	86.57 $\pm$ 0.19	95.28 $\pm$ 1.80
	EMKMC	82.75 $\pm$ 0.07	95.64 $\pm$ 1.41	83.56 $\pm$ 0.80	84.35 $\pm$ 0.68	98.41 $\pm$ 1.43	97.12 $\pm$ 0.64	73.56 $\pm$ 1.87
	AEVC	86.87 $\pm$ 1.37	94.88 $\pm$ 0.29	85.86 $\pm$ 1.68	86.61 $\pm$ 0.92	89.91 $\pm$ 1.70	81.85 $\pm$ 1.09	92.17 $\pm$ 1.67
	3AMVC	87.68 $\pm$ 0.70	96.57 $\pm$ 1.94	87.99 $\pm$ 1.24	88.63 $\pm$ 0.13	91.04 $\pm$ 0.01	82.98 $\pm$ 1.69	95.41 $\pm$ 0.16
	SMVAGC-SF	86.19 $\pm$ 1.22	96.39 $\pm$ 0.56	86.84 $\pm$ 0.67	87.56 $\pm$ 0.55	89.75 $\pm$ 1.15	81.12 $\pm$ 1.60	95.03 $\pm$ 0.42
	ALPC	80.28 $\pm$ 0.88	85.63 $\pm$ 0.07	74.78 $\pm$ 0.51	76.11 $\pm$ 0.53	82.64 $\pm$ 1.65	71.05 $\pm$ 0.73	81.96 $\pm$ 0.73
	OMVCDR	62.43 $\pm$ 0.00	82.71 $\pm$ 0.00	61.46 $\pm$ 0.00	63.87 $\pm$ 0.00	70.00 $\pm$ 0.00	50.33 $\pm$ 0.00	87.36 $\pm$ 0.00
	Ours	98.01 $\pm$ 1.28	98.76 $\pm$ 0.78	97.26 $\pm$ 2.01	97.15 $\pm$ 1.71	98.95 $\pm$ 0.71	98.25 $\pm$ 1.11	98.31 $\pm$ 1.09
Cal20	FMCNOF	54.11 $\pm$ 1.78	48.41 $\pm$ 1.91	42.66 $\pm$ 0.03	51.62 $\pm$ 1.09	64.56 $\pm$ 1.76	53.73 $\pm$ 1.35	49.68 $\pm$ 1.15
	EOMSC-CA	53.02 $\pm$ 0.01	49.59 $\pm$ 0.96	43.78 $\pm$ 1.57	52.20 $\pm$ 1.98	67.39 $\pm$ 1.06	56.71 $\pm$ 0.79	48.36 $\pm$ 0.63
	FMVACC	42.60 $\pm$ 0.81	55.06 $\pm$ 0.05	28.65 $\pm$ 0.20	34.52 $\pm$ 0.18	71.13 $\pm$ 0.04	62.20 $\pm$ 0.66	23.11 $\pm$ 1.05
	EMKMC	52.23 $\pm$ 0.71	34.64 $\pm$ 1.61	43.24 $\pm$ 1.38	65.65 $\pm$ 0.13	43.76 $\pm$ 0.73	32.43 $\pm$ 0.52	32.24 $\pm$ 0.26
	AEVC	42.45 $\pm$ 1.57	57.12 $\pm$ 1.81	29.31 $\pm$ 0.06	34.89 $\pm$ 0.22	70.18 $\pm$ 0.45	62.54 $\pm$ 0.20	23.47 $\pm$ 0.58
	3AMVC	42.23 $\pm$ 1.63	57.21 $\pm$ 1.65	27.45 $\pm$ 0.97	34.35 $\pm$ 0.49	70.12 $\pm$ 0.81	61.12 $\pm$ 1.36	23.31 $\pm$ 0.39
	SMVAGC-SF	32.56 $\pm$ 1.50	44.25 $\pm$ 0.16	22.39 $\pm$ 1.58	28.57 $\pm$ 1.82	64.12 $\pm$ 0.92	58.37 $\pm$ 1.50	18.92 $\pm$ 0.67
	ALPC	29.25 $\pm$ 0.27	30.70 $\pm$ 0.88	17.64 $\pm$ 0.98	25.46 $\pm$ 0.52	54.69 $\pm$ 0.47	43.65 $\pm$ 0.92	17.97 $\pm$ 0.83
	OMVCDR	41.91 $\pm$ 0.00	39.18 $\pm$ 0.00	33.67 $\pm$ 0.00	42.24 $\pm$ 0.00	58.72 $\pm$ 0.00	53.19 $\pm$ 0.00	35.04 $\pm$ 0.00
	Ours	63.41 $\pm$ 0.48	59.74 $\pm$ 0.39	67.34 $\pm$ 1.11	71.90 $\pm$ 1.61	73.60 $\pm$ 1.33	83.65 $\pm$ 0.37	63.05 $\pm$ 1.49
YTF10	FMCNOF	70.21 $\pm$ 0.62	76.34 $\pm$ 0.67	58.72 $\pm$ 1.21	64.63 $\pm$ 0.89	72.46 $\pm$ 1.62	59.76 $\pm$ 1.93	71.78 $\pm$ 0.07
	EOMSC-CA	73.17 $\pm$ 1.62	76.42 $\pm$ 0.62	58.67 $\pm$ 0.63	64.54 $\pm$ 0.34	74.36 $\pm$ 0.17	58.77 $\pm$ 0.96	71.78 $\pm$ 0.61
	FMVACC	76.64 $\pm$ 1.47	77.12 $\pm$ 0.24	68.90 $\pm$ 1.02	72.28 $\pm$ 1.23	78.89 $\pm$ 0.51	72.77 $\pm$ 1.31	71.81 $\pm$ 1.98
	EMKMC	57.91 $\pm$ 1.25	69.45 $\pm$ 0.39	45.97 $\pm$ 1.47	49.34 $\pm$ 0.85	69.35 $\pm$ 0.27	56.31 $\pm$ 0.94	43.91 $\pm$ 1.82
	AEVC	70.93 $\pm$ 0.44	75.55 $\pm$ 0.59	62.63 $\pm$ 1.53	66.74 $\pm$ 1.33	74.93 $\pm$ 1.63	66.41 $\pm$ 0.05	67.13 $\pm$ 0.99
	3AMVC	61.89 $\pm$ 1.01	69.93 $\pm$ 0.39	45.79 $\pm$ 0.62	52.55 $\pm$ 0.85	68.90 $\pm$ 0.15	46.62 $\pm$ 0.56	60.69 $\pm$ 0.32
	SMVAGC-SF	76.54 $\pm$ 1.75	79.01 $\pm$ 0.99	71.32 $\pm$ 0.24	72.41 $\pm$ 0.99	78.05 $\pm$ 1.41	73.98 $\pm$ 1.36	67.21 $\pm$ 1.24
	ALPC	75.85 $\pm$ 0.38	71.96 $\pm$ 0.19	60.01 $\pm$ 0.02	64.52 $\pm$ 1.25	75.87 $\pm$ 1.98	63.67 $\pm$ 0.93	66.84 $\pm$ 0.78
	OMVCDR	70.17 $\pm$ 0.00	75.35 $\pm$ 0.00	67.04 $\pm$ 0.00	70.66 $\pm$ 0.00	78.31 $\pm$ 0.00	70.20 $\pm$ 0.00	71.14 $\pm$ 0.00
	Ours	83.11 $\pm$ 2.32	79.98 $\pm$ 1.17	71.62 $\pm$ 1.29	74.77 $\pm$ 1.57	83.11 $\pm$ 0.92	74.20 $\pm$ 1.22	76.09 $\pm$ 0.52
MNIST	FMCNOF	46.61 $\pm$ 0.23	52.19 $\pm$ 0.05	34.09 $\pm$ 1.13	42.28 $\pm$ 1.42	46.74 $\pm$ 0.85	34.12 $\pm$ 0.16	55.60 $\pm$ 1.84
	EOMSC-CA	98.00 $\pm$ 1.61	94.59 $\pm$ 0.01	95.62 $\pm$ 1.42	96.06 $\pm$ 1.85	98.00 $\pm$ 1.62	96.03 $\pm$ 0.00	96.10 $\pm$ 0.95
	FMVACC	97.78 $\pm$ 0.56	96.38 $\pm$ 0.89	96.57 $\pm$ 1.77	96.92 $\pm$ 1.87	98.15 $\pm$ 0.45	96.53 $\pm$ 0.22	97.37 $\pm$ 1.32
	EMKMC	89.00 $\pm$ 0.62	93.00 $\pm$ 1.04	88.00 $\pm$ 1.11	89.00 $\pm$ 0.51	98.00 $\pm$ 0.22	97.00 $\pm$ 1.59	82.00 $\pm$ 1.60
	AEVC	96.46 $\pm$ 1.97	94.91 $\pm$ 0.83	94.65 $\pm$ 0.14	95.21 $\pm$ 1.36	96.76 $\pm$ 0.98	94.11 $\pm$ 0.88	96.47 $\pm$ 1.57
	3AMVC	98.38 $\pm$ 0.44	96.56 $\pm$ 0.64	97.03 $\pm$ 0.25	97.33 $\pm$ 0.93	98.49 $\pm$ 1.73	97.19 $\pm$ 1.11	97.50 $\pm$ 0.45
	SMVAGC-SF	82.96 $\pm$ 1.66	85.40 $\pm$ 1.82	77.85 $\pm$ 1.26	80.27 $\pm$ 1.71	85.17 $\pm$ 0.95	74.74 $\pm$ 1.16	87.05 $\pm$ 1.33
	ALPC	96.43 $\pm$ 1.83	91.44 $\pm$ 1.65	92.32 $\pm$ 0.31	93.09 $\pm$ 1.03	96.43 $\pm$ 1.39	93.04 $\pm$ 1.02	93.14 $\pm$ 0.77
	OMVCDR	83.32 $\pm$ 0.00	85.77 $\pm$ 0.00	81.25 $\pm$ 0.00	83.25 $\pm$ 0.00	87.27 $\pm$ 0.00	78.21 $\pm$ 0.00	89.00 $\pm$ 0.00
	Ours	99.05 $\pm$ 1.68	97.05 $\pm$ 0.13	97.91 $\pm$ 0.84	98.12 $\pm$ 1.25	99.05 $\pm$ 1.64	98.13 $\pm$ 0.49	98.11 $\pm$ 1.43
Cifar10	FMCNOF	20.34 $\pm$ 1.78	12.66 $\pm$ 0.81	7.70 $\pm$ 0.33	21.14 $\pm$ 1.97	20.47 $\pm$ 1.17	14.51 $\pm$ 1.05	20.92 $\pm$ 0.68
	EOMSC-CA	25.16 $\pm$ 0.39	13.90 $\pm$ 0.40	8.16 $\pm$ 1.37	17.25 $\pm$ 0.30	26.28 $\pm$ 0.79	16.75 $\pm$ 1.51	21.55 $\pm$ 0.37
	FMVACC	25.68 $\pm$ 0.70	14.81 $\pm$ 0.69	9.10 $\pm$ 0.62	18.23 $\pm$ 1.17	28.01 $\pm$ 0.63	18.15 $\pm$ 0.95	18.30 $\pm$ 0.68
	EMKMC	21.12 $\pm$ 0.41	7.02 $\pm$ 0.15	4.02 $\pm$ 0.84	14.23 $\pm$ 1.81	24.21 $\pm$ 0.08	14.02 $\pm$ 0.35	13.51 $\pm$ 0.33
	AEVC	25.92 $\pm$ 0.72	12.69 $\pm$ 0.92	7.68 $\pm$ 0.72	16.97 $\pm$ 0.76	28.35 $\pm$ 0.34	16.85 $\pm$ 0.98	17.10 $\pm$ 0.48
	3AMVC	25.01 $\pm$ 0.90	13.57 $\pm$ 0.90	8.09 $\pm$ 0.32	18.12 $\pm$ 1.00	28.12 $\pm$ 0.64	17.34 $\pm$ 0.66	18.21 $\pm$ 0.41
	SMVAGC-SF	24.46 $\pm$ 0.60	12.83 $\pm$ 0.41	7.55 $\pm$ 0.65	16.82 $\pm$ 0.95	27.52 $\pm$ 0.16	16.77 $\pm$ 0.83	16.87 $\pm$ 0.73
	ALPC	17.77 $\pm$ 1.35	4.43 $\pm$ 0.36	2.25 $\pm$ 1.67	13.16 $\pm$ 1.89	18.62 $\pm$ 0.54	11.79 $\pm$ 1.18	14.89 $\pm$ 0.88
	OMVCDR	15.23 $\pm$ 0.00	11.14 $\pm$ 0.00	8.24 $\pm$ 0.00	16.32 $\pm$ 0.00	14.21 $\pm$ 0.00	15.21 $\pm$ 0.00	17.32 $\pm$ 0.00
	Ours	27.01 $\pm$ 0.54	14.92 $\pm$ 0.13	9.24 $\pm$ 0.31	18.41 $\pm$ 0.91	28.67 $\pm$ 0.26	17.06 $\pm$ 1.80	21.78 $\pm$ 0.87
YTF20	FMCNOF	22.68 $\pm$ 0.10	29.12 $\pm$ 1.54	12.77 $\pm$ 1.57	20.77 $\pm$ 1.18	22.69 $\pm$ 1.08	12.59 $\pm$ 1.65	59.38 $\pm$ 0.72
	EOMSC-CA	72.37 $\pm$ 0.64	75.23 $\pm$ 1.61	63.30 $\pm$ 1.76	65.39 $\pm$ 1.95	75.08 $\pm$ 1.75	63.65 $\pm$ 1.73	65.32 $\pm$ 0.32
	FMVACC	71.43 $\pm$ 0.63	75.18 $\pm$ 1.40	63.82 $\pm$ 1.62	65.84 $\pm$ 1.25	75.12 $\pm$ 0.36	65.03 $\pm$ 1.41	65.01 $\pm$ 1.97
	EMKMC	58.23 $\pm$ 0.65	71.21 $\pm$ 0.41	47.67 $\pm$ 0.50	52.28 $\pm$ 0.86	69.21 $\pm$ 0.35	58.42 $\pm$ 1.26	47.91 $\pm$ 1.78
	AEVC	69.12 $\pm$ 1.75	77.89 $\pm$ 1.10	61.48 $\pm$ 1.08	63.73 $\pm$ 1.41	74.59 $\pm$ 1.23	60.89 $\pm$ 1.79	67.02 $\pm$ 0.06
	3AMVC	63.15 $\pm$ 1.76	74.64 $\pm$ 0.88	40.46 $\pm$ 1.58	44.58 $\pm$ 0.50	69.19 $\pm$ 0.92	36.14 $\pm$ 1.95	58.77 $\pm$ 0.64
	SMVAGC-SF	69.42 $\pm$ 0.85	75.15 $\pm$ 1.30	61.68 $\pm$ 1.60	63.88 $\pm$ 0.91	74.97 $\pm$ 1.56	61.72 $\pm$ 1.59	66.28 $\pm$ 1.62
	ALPC	65.46 $\pm$ 0.91	68.64 $\pm$ 1.04	53.24 $\pm$ 0.39	55.86 $\pm$ 1.06	69.22 $\pm$ 1.60	56.18 $\pm$ 1.12	56.25 $\pm$ 0.98
	OMVCDR	63.83 $\pm$ 0.00	72.79 $\pm$ 0.00	55.08 $\pm$ 0.00	57.91 $\pm$ 0.00	70.26 $\pm$ 0.00	51.07 $\pm$ 0.00	66.87 $\pm$ 0.00
	Ours	75.39 $\pm$ 1.07	77.01 $\pm$ 1.54	65.71 $\pm$ 0.94	67.61 $\pm$ 0.29	77.24 $\pm$ 1.90	67.60 $\pm$ 1.15	67.62 $\pm$ 1.97
YTF50	FMCNOF	70.48 $\pm$ 0.63	83.36 $\pm$ 1.03	60.50 $\pm$ 1.79	61.49 $\pm$ 1.43	73.83 $\pm$ 0.03	54.67 $\pm$ 1.79	70.26 $\pm$ 1.99
	EOMSC-CA	69.71 $\pm$ 1.95	83.22 $\pm$ 0.62	60.23 $\pm$ 1.94	63.12 $\pm$ 1.29	70.58 $\pm$ 0.18	60.18 $\pm$ 0.12	66.22 $\pm$ 0.12
	FMVACC	70.09 $\pm$ 1.57	83.49 $\pm$ 0.84	62.08 $\pm$ 1.52	62.96 $\pm$ 1.93	74.11 $\pm$ 1.89	60.64 $\pm$ 1.59	65.52 $\pm$ 0.39
	EMKMC	61.01 $\pm$ 1.81	77.27 $\pm$ 0.02	51.34 $\pm$ 0.19	52.55 $\pm$ 0.44	72.64 $\pm$ 1.02	59.01 $\pm$ 1.46	47.31 $\pm$ 0.44
	AEVC	70.13 $\pm$ 0.78	83.60 $\pm$ 0.12	62.63 $\pm$ 0.27	63.51 $\pm$ 0.06	76.20 $\pm$ 0.28	60.82 $\pm$ 1.15	66.50 $\pm$ 0.64
	3AMVC	67.61 $\pm$ 1.33	82.57 $\pm$ 1.08	43.08 $\pm$ 1.12	44.75 $\pm$ 1.32	74.36 $\pm$ 0.21	34.68 $\pm$ 1.12	63.80 $\pm$ 0.63
	SMVAGC-SF	68.29 $\pm$ 1.76	84.09 $\pm$ 0.82	62.50 $\pm$ 1.76	63.41 $\pm$ 1.13	74.48 $\pm$ 0.96	59.33 $\pm$ 1.66	68.26 $\pm$ 0.67
	ALPC	41.49 $\pm$ 1.25	51.19 $\pm$ 0.25	24.57 $\pm$ 1.27	26.29 $\pm$ 0.08	45.82 <		

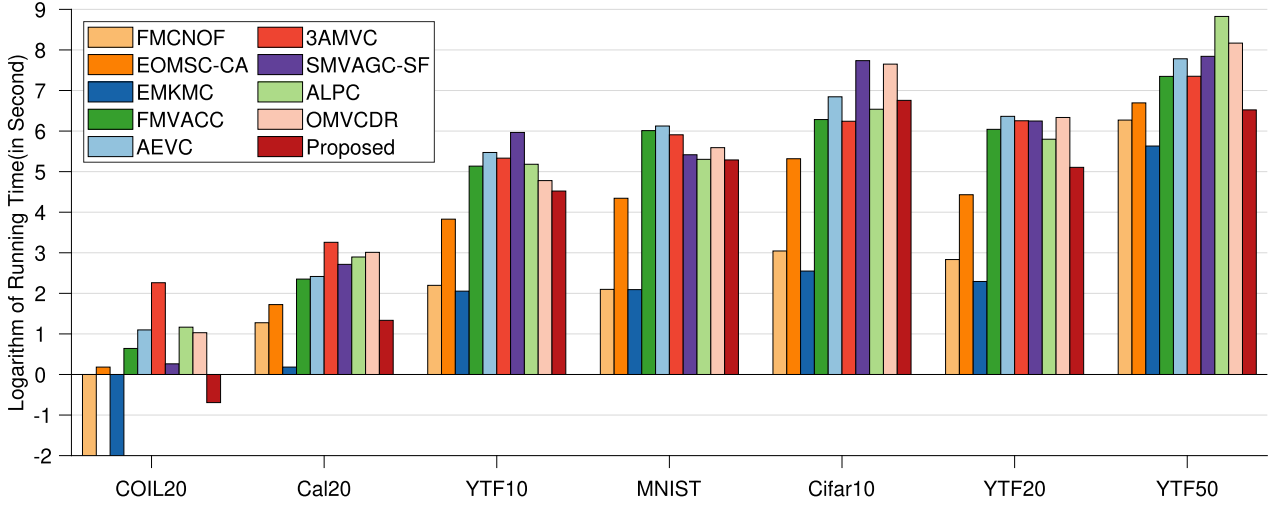


Fig. 4. Logarithm of the running time for different algorithms on each benchmark dataset. Negative values indicate runtimes below one second.

playing a critical role in improving clustering performance. To assess their individual contributions, we conducted ablation studies on seven benchmark datasets. Specifically, we designed four variants of TCAA, as described below.

(1) *Effect of all three consistency strategies*: To evaluate the overall impact of integrating all three consistency strategies, we directly linearly combine the anchor graphs learned from (3) to obtain a consensus anchor graph $\mathbf{Z}$ (i.e., $\mathbf{Z} = \sum_{t=1}^v \mathbf{Z}^{(t)}/v$), followed by spectral clustering on this consensus graph. We denote this variant as ‘w/o All’.

(2) *Effect of the spectral consistency strategy*: Spectral consistency exploits spectral anchor signatures to capture the global structural roles of anchors, thereby improving robustness to noise and mitigating the effect of ambiguous local similarities. It is particularly beneficial in cases where anchors from different views cannot be reliably aligned by representation consistency. To evaluate its contribution, we remove this term by setting $\beta = 0$ in (8) and denote this variant as ‘w/o Spectral’.

(3) *Effect of the discrete clustering consistency strategy*: To investigate the effect of discrete clustering consistency, we eliminate this component by setting $\lambda = 0$ in (8). The clustering result is then obtained by applying spectral clustering to the consensus graph. We denote the resulting variant as ‘w/o Discrete’.

(4) *Effect of adaptive view weighting*: To verify the benefit of adaptive weighting in suppressing noisy or less informative views, we further consider a variant with fixed uniform view weights, i.e., $\alpha^{(t)} = 1$, $t = 1, \dots, v$. We denote this variant as ‘w/o Weight’.

The experimental results of the four variants are reported in Table IV. Based on the results, we can see that removing all consistency modules (‘w/o All’) yields the poorest performance on all datasets. In particular, on datasets Caltech101-20 and MNIST, compared to the proposed method, the accuracy reduces by 20% and 16%, respectively, underscoring the necessity of these consistency strategies. Excluding the spectral consistency mechanism (‘w/o Spectral’) results in a noticeable performance

TABLE IV
ABLATION STUDY RESULTS IN TERMS OF ACC, NMI, AND ARI ON BENCHMARK DATASETS

Variants	COIL20	Cal20	MNIST	Cifar10	YTF10	YTF20	YTF50
ACC							
w/o All	84.83	43.74	82.72	20.33	71.69	68.15	65.36
w/o Spectral	85.91	62.36	98.33	21.63	76.53	72.19	67.57
w/o Discrete	87.37	59.68	98.03	26.64	80.73	72.97	68.14
w/o Weight	82.43	36.92	98.94	23.18	80.53	67.14	71.23
Proposed	98.01	63.41	99.05	27.01	83.11	75.39	73.71
NMI							
w/o All	95.42	52.38	87.62	11.62	79.68	76.09	81.25
w/o Spectral	96.52	59.56	95.25	10.77	76.76	76.77	82.43
w/o Discrete	96.65	54.21	96.99	13.55	78.51	76.84	68.14
w/o Weight	92.23	54.81	96.81	9.13	78.09	77.82	83.42
Proposed	98.76	59.74	97.05	14.92	79.98	77.97	84.82
ARI							
w/o All	85.72	33.29	84.41	6.46	69.98	61.01	61.94
w/o Spectral	87.68	67.29	96.32	5.22	65.49	65.48	59.89
w/o Discrete	87.47	48.87	97.87	7.58	65.83	62.63	63.21
w/o Weight	80.74	32.66	97.67	5.22	69.14	60.40	57.74
Proposed	97.26	67.34	97.91	9.24	71.62	65.71	63.32

degradation compared with the proposed model, indicating that spectral consistency plays a crucial role in enhancing the cluster structure. Similarly, the removal of the discrete clustering constraint (‘w/o Discrete’) also reduces performance, particularly on datasets such as COIL20 and YouTubeFace50, where the accuracy reduces by 11% and 5.6%, respectively, demonstrating the effectiveness of incorporating discrete cluster assignments. By contrast, the proposed model consistently achieves the best results on all datasets, confirming that the joint integration of three consistency mechanisms leads to superior clustering performance. Compared with the fixed-weight variant, the proposed model consistently achieves better performance, which confirms that adaptively learned view weights help reduce the adverse effect of noisy or weakly informative views.

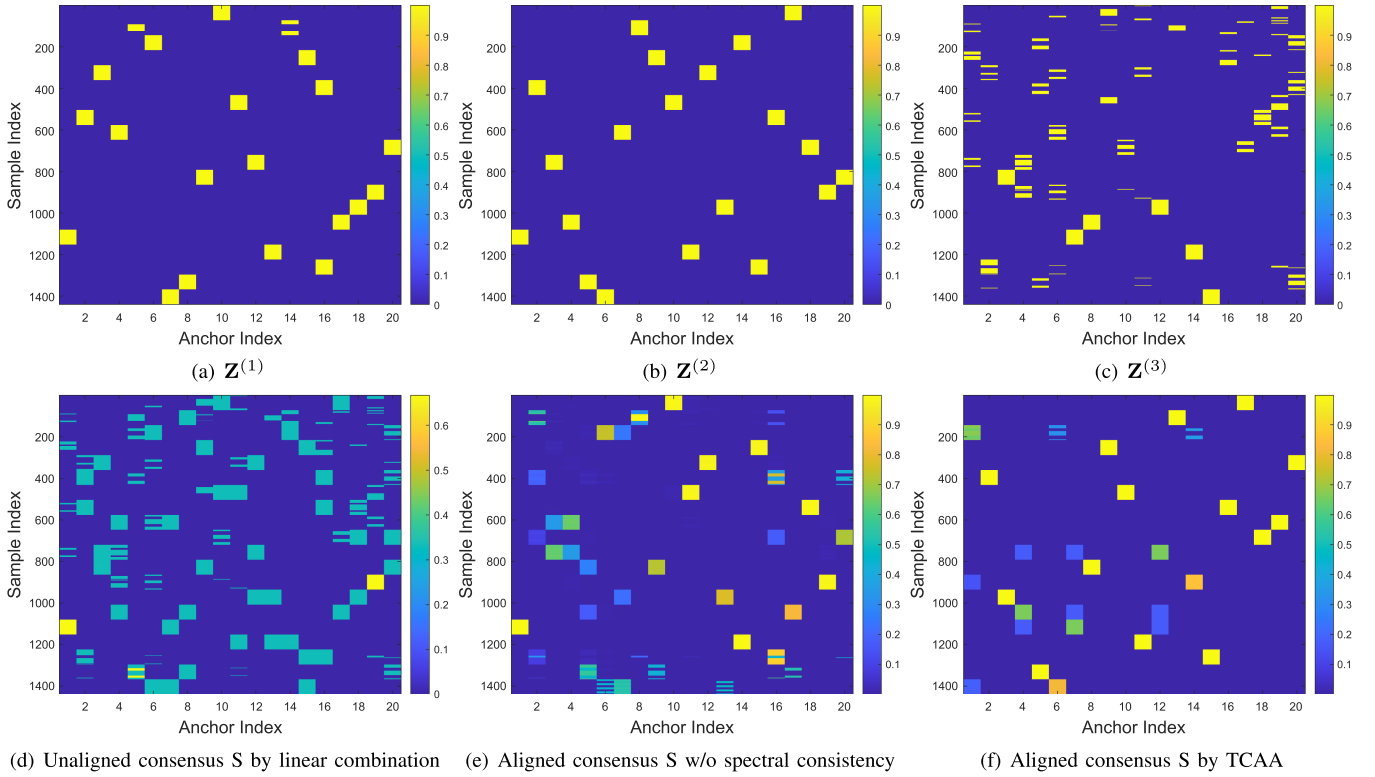


Fig. 5. Visualization of anchor graphs on COIL20. (a)–(c): anchor graphs of individual views; (d): unaligned consensus anchor graph; (e): aligned consensus anchor graph without spectral consistency; (f): aligned consensus anchor graph obtained by our three-consistency mechanism.

E. Anchor Graph Visualization

We visualize the effectiveness of the proposed alignment mechanism on the COIL20 dataset in Fig. 5. For clearer comparison, the number of anchors connected to each sample is set to 1 when constructing the anchor graphs for each view. Subfigures (a) – (c) show the learned anchor graphs $\mathbf{Z}^{(1)}$, $\mathbf{Z}^{(2)}$, and $\mathbf{Z}^{(3)}$ for the three views, where each row corresponds to a sample and each column represents an anchor. The bright yellow blocks indicate high similarity or strong connections between samples and their corresponding anchors, revealing that each view independently forms distinct yet view-specific local structures. However, the spatial positions of these bright regions differ across views, demonstrating a significant AUP. Subfigure (d) presents the unaligned consensus anchor graph $\mathbf{S}$ obtained through simple linear averaging of the individual anchor graphs. The scattered and blurred block boundaries indicate that direct fusion fails to maintain consistent cluster structures. Subfigure (e) illustrates the aligned consensus anchor graph $\mathbf{S}$ without spectral consistency, which partially alleviates misalignment but still exhibits fragmented and noisy regions, suggesting incomplete correspondence correction. Finally, subfigure (f) shows the aligned consensus anchor graph $\mathbf{S}$ generated by the proposed TCAA model, where both representation and spectral consistencies are enforced. The resulting matrix displays clear block-diagonal patterns and minimal cross-cluster interference, demonstrating that the anchors across views have

been successfully aligned and integrated into a coherent global structure.

F. Anchor Selection Analysis

To assess the effect of anchor selection on TCAA, we further compare four commonly used anchor initialization strategies, namely Random Sampling, DAS, BKHK, and k -means, while keeping all other settings unchanged. The results in Table V, measured by ACC, NMI, and ARI, show that TCAA remains competitive under all four strategies, indicating that the proposed framework is not sensitive to a particular anchor initialization method. Among them, k -means achieves the best overall performance. In particular, it yields the best ACC on six of the seven datasets, and consistently attains the best NMI and ARI on all seven datasets. This suggests that the anchors produced by k -means are more representative and discriminative, thereby benefiting the subsequent consensus learning and cross-view alignment. Therefore, although TCAA is compatible with different anchor selection strategies, we use k -means as the default choice in Algorithm 1 due to its superior and more stable empirical performance.

G. Convergence Analysis

In the optimization produces, each variable block is optimized while keeping the others fixed, and every subproblem admits an exact or closed-form solution. This design ensures that

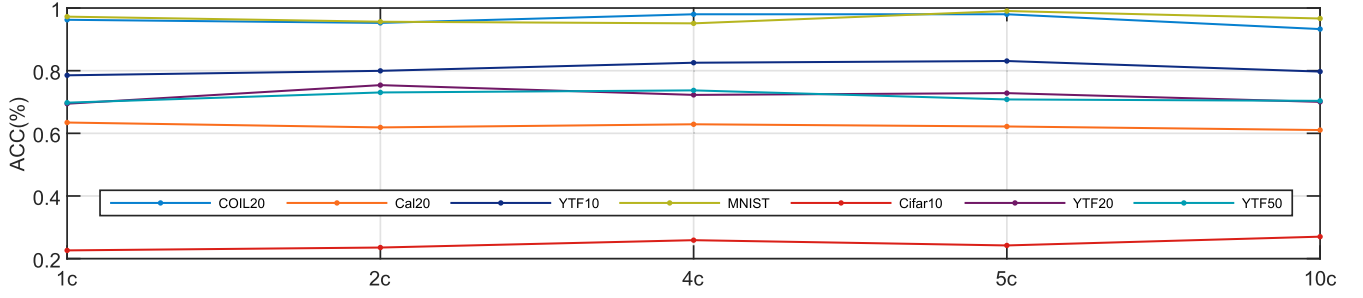


Fig. 6. Sensitivity study for anchor number m on different datasets. The horizontal axis represents the ratio between the number of anchors m and the number of clusters c . The vertical axis denotes the clustering accuracy obtained on various benchmark datasets.

TABLE V
CLUSTERING RESULTS UNDER DIFFERENT ANCHOR SELECTION STRATEGIES

Strategies	COIL20	Cal20	MNIST	Cifar10	YTF10	YTF20	YTF50
ACC							
Random	97.57	47.40	93.96	20.87	80.51	72.80	70.19
DAS	98.06	46.69	98.53	21.25	62.10	52.40	50.28
BKHK	94.86	53.19	98.85	23.70	76.74	69.48	68.88
k -means	98.01	63.41	99.05	27.01	83.11	75.39	73.71
NMI							
Random	98.27	49.07	89.55	8.27	79.48	76.25	77.95
DAS	98.75	47.66	95.85	7.75	59.09	54.60	61.15
BKHK	98.72	56.78	96.52	11.59	78.49	75.03	80.45
k -means	98.76	62.98	97.05	14.92	79.98	78.97	84.82
ARI							
Random	95.92	46.62	88.13	4.20	68.75	63.72	56.96
DAS	97.12	40.02	96.78	4.50	49.73	33.96	38.23
BKHK	94.77	54.09	97.47	6.08	69.39	58.50	61.23
k -means	97.26	67.34	97.91	9.24	71.62	65.71	63.32

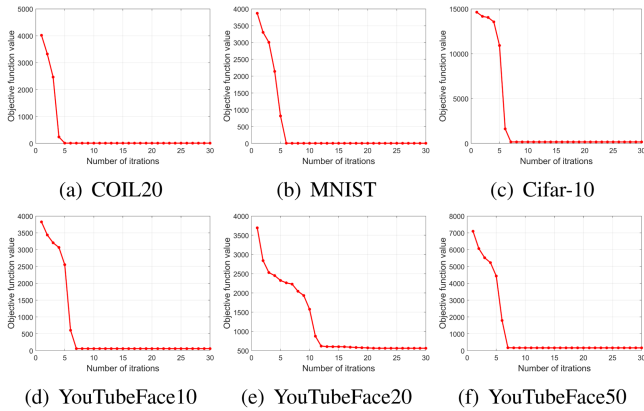


Fig. 7. The convergence curves of our algorithm on six datasets.

the overall objective function is monotonically non-increasing. Moreover, the objective value of (8) is bounded below by zero. These two properties together guarantee the convergence of the objective sequence. Fig. 7 illustrates the convergence behavior of the proposed method. Each subfigure plots the number of iterations on the horizontal axis and the corresponding objective values on the vertical axis. As observed, the objective function decreases rapidly and monotonically across all datasets. Except

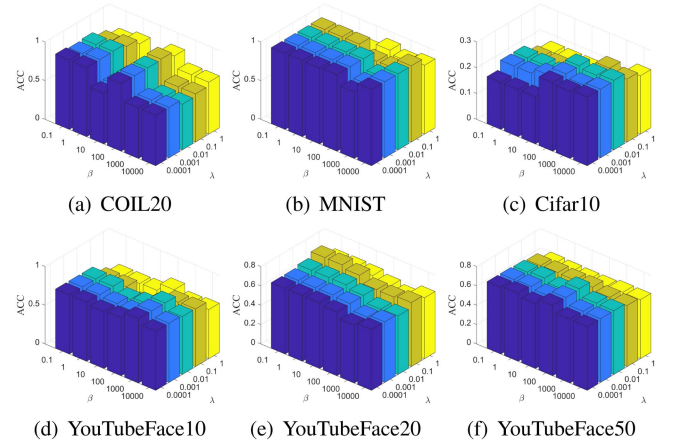


Fig. 8. The sensitivity studies of hyper-parameters λ and β on six datasets.

for YouTubeFace20, convergence is achieved within 10 iterations, demonstrating the efficiency and fast convergence of the proposed approach.

H. Parameter Analysis

TCAA involves two hyperparameters, namely β and λ , which control the contributions of spectral consistency and discrete clustering consistency, respectively. Fig. 8 reports the variation of ACC with respect to these two parameters across six benchmark datasets with the anchor number fixed. As shown in the figure, except for YouTubeFace20, the proposed model achieves optimal clustering performance when β is set to relatively large values (e.g., $\beta = 100$ or $\beta = 1000$). This finding highlights the importance of the spectral consistency mechanism in our framework. Moreover, Fig. 8 shows that smaller values of λ generally lead to better clustering results on most datasets, suggesting that a moderate regularization on discrete clustering consistency is more beneficial.

To further investigate the impact of anchor number on clustering performance, we tested $m = 1c, 2c, 4c, 5c$, and $10c$, while performing grid searches on the remaining parameters to record the optimal ACC metric. As shown in Fig. 6, the model is generally insensitive to the number of anchors, and clustering performance remains relatively stable as m varies. This robustness demonstrates that the proposed approach can achieve

consistent clustering performance even under different anchor settings.

VI. CONCLUSION

In this study, we proposed a novel and efficient anchor-based multi-view clustering method, named TCAA, to address the AUP in multi-view clustering. Unlike methods that align all views to a fixed reference, TCAA adaptively balances view contributions and jointly learns an aligned consensus anchor graph that integrates both representation and spectral consistency. By iteratively updating permutation matrices and view weights, it suppresses noisy or unreliable views while emphasizing informative ones, thus eliminating reference-view bias and achieving robust alignment. Furthermore, clustering consistency is enforced by directly aligning the consensus anchor graph with discrete cluster assignments, unifying graph learning and clustering in a single framework, and reducing information loss compared with traditional two-stage methods.

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Chuan Tang received the MS degree in computer science and technology from China University of Geosciences, Wuhan, China, in 2024. He is currently working toward the PhD degree with the National University of Defense Technology, China. He has authored or coauthored several peer-reviewed papers, such as *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Neural Networks and Learning Systems*, and *Neural Networks*. His research interests include multi-view learning.



Miaomiao Li received the PhD degree from National University of Defense Technology, China. She is currently an associate professor with Changsha University, Changsha, China. She has authored or coauthored several peer-reviewed papers, such as *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IEEE Transactions on Neural Networks and Learning Systems*, *AAAI*, *IJCAI*, and *Neurocomputing*. Her research interests include kernel learning and multi-view clustering. She serves on the Technical Program Committees of IJCAI 2017–2022.



Jun Wang received the MS degree from China University of Geosciences, Wuhan, China, in 2023. He is currently working toward the PhD degree with the School of Computer Science, National University of Defense Technology. He has authored or coauthored several peer-reviewed papers, such as *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Cybernetics*, *Neural Networks*, and *Information Fusion*. His research focuses on multi-view learning.



Renxiang Guan received the BS degree from China University of Geosciences (CUG), Wuhan, China in 2023. He is currently working toward the master's degree with the National University of Defense Technology (NUDT), Changsha, China. He has authored or coauthored several peer-reviewed papers, including *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Geoscience and Remote Sensing*, *AAAI*, *ICASSP*, *Knowledge-Based Systems*, *IJCNN*, and *Remote Sensing*. His research interests include deep graph clustering, and hyperspectral image processing.



Siwei Wang (Member, IEEE) received the PhD degree from the National University of Defense Technology (NUDT), China. He is currently an assistant research Professor with the Intelligent Game and Decision Lab, Academy of Military Science, China. He has authored or coauthored several papers and served as an AC in top journals and conferences, such as *ICML/NeurIPS/CVPR/AAAI/ICCV*, *IEEE Transactions on Pattern Analysis and Machine Intelligence/IEEE Transactions on Image Processing/IEEE Transactions on Knowledge and Data Engineering* and *IEEE Transactions on Neural Networks and Learning Systems*. His research interests include unsupervised multi-modal learning and scalable clustering. He serves as the associate editor of *Pattern Recognition Journal*.



Chang Tang (Senior Member, IEEE) received the PhD degree from Tianjin University in 2016. He is currently an associate professor with the Huazhong University of Science and Technology, Wuhan, China. He has authored or coauthored more than 100 peer-reviewed papers, including those in highly regarded journals and conferences such as *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Knowledge and Data Engineering*, ICCV, CVPR, IJCAI, AAAI, and ACM MM. His research interests

include machine learning and CV. He serves as an associate editor of *Neural Networks*, *IEEE Journal of Biomedical and Health Informatics*, *CAAI Transactions on Intelligence Technology* and *BMC Bioinformatics*. He serves as technical program committee or as area chair of some top conferences such as NIPS, ICML, CVPR, ICCV, ECCV, IJCAI, ICME and AAAI. He won the best paper award at the 5th Asian Conference on Artificial Intelligence Technology.



Xinwang Liu (Senior Member, IEEE) received the PhD degree from National University of Defense Technology (NUDT), China. He is currently a full professor with the School of Computer, NUDT. He has authored or coauthored more than 60 peer-reviewed papers, including those in highly regarded journals and conferences such as *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Image Processing*, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Information Forensics and Security*, *NeurIPS*, ICCV, CVPR, AAAI, and IJCAI. His research

interests include kernel learning and unsupervised feature learning. More information can be found at xinwangliu.github.io.



En Zhu received the PhD degree from National University of Defense Technology (NUDT), China. He is currently a professor with the School of Computer Science, NUDT, China. He has authored or coauthored more than 150 peer-reviewed papers, including *IEEE Transactions on Circuits and Systems for Video Technology*, *IEEE Transactions on Neural Networks and Learning Systems*, PR, AAAI, and IJCAI. His research interests include pattern recognition, image processing, machine vision and machine learning. He was awarded the China National Excellence Doctoral

Dissertation.